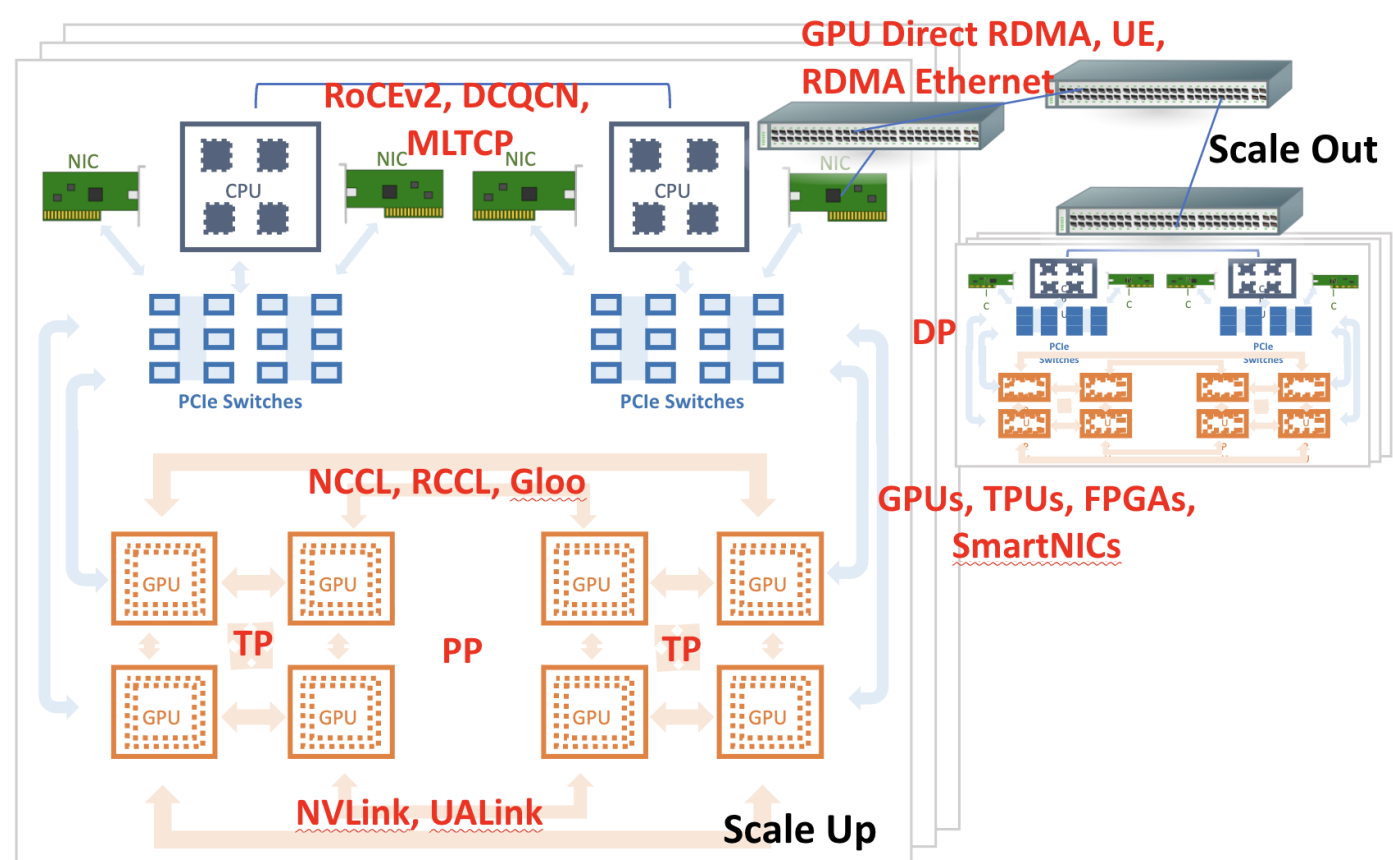


The Problem

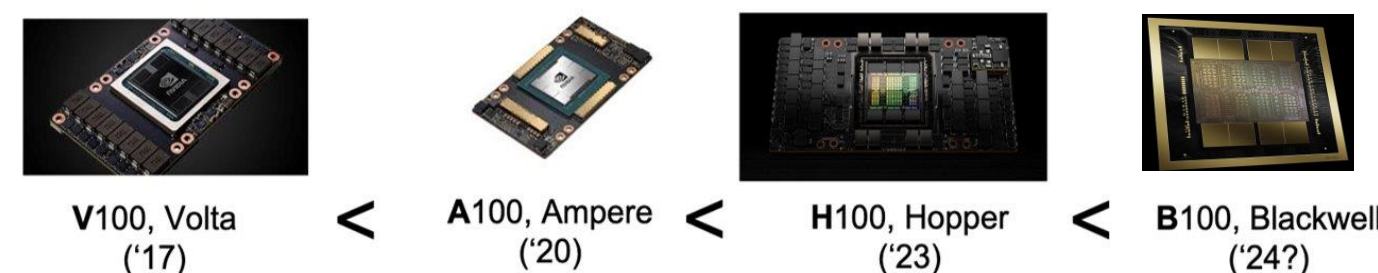
Complex AI training stack

Llama 4 $\rightarrow \sim 32$ k NVIDIA H100 GPUs



Using physical infrastructure to test configurations is **EXPENSIVE!**
LLM simulators AstraSim^[7], SimAI^[8] predict performance!

Why Heterogeneity

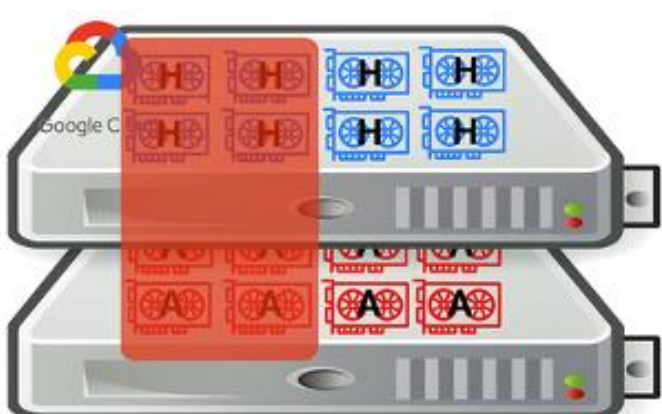
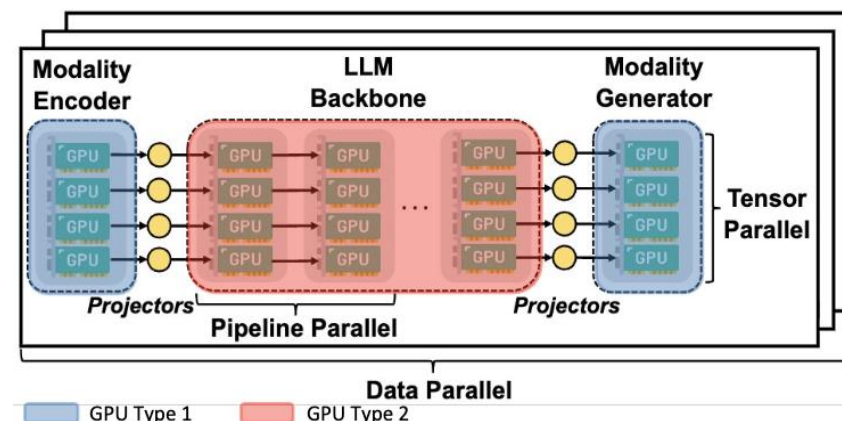


Compute & networking hardware changes every ~ 2 years



Modern AI clusters span multiple data centers due to power demands in the tens of megawatts!

Leveraging device heterogeneity for improved utilization in multimodal training!

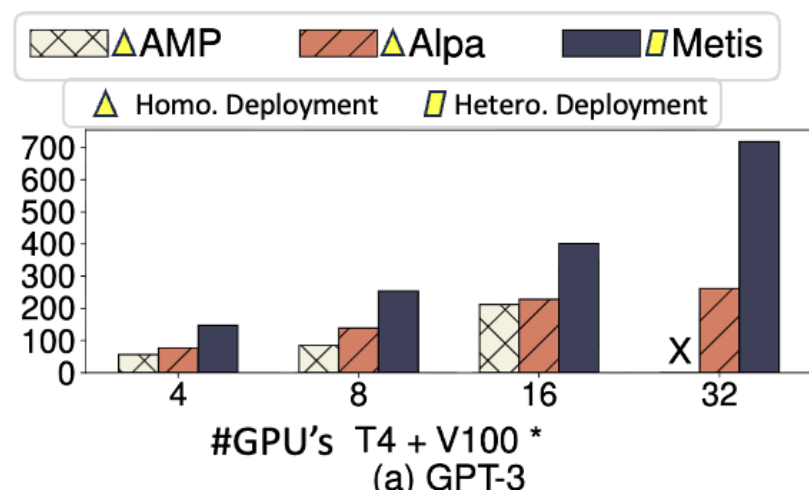


Heterogeneous deployment of the model in cloud infrastructure!

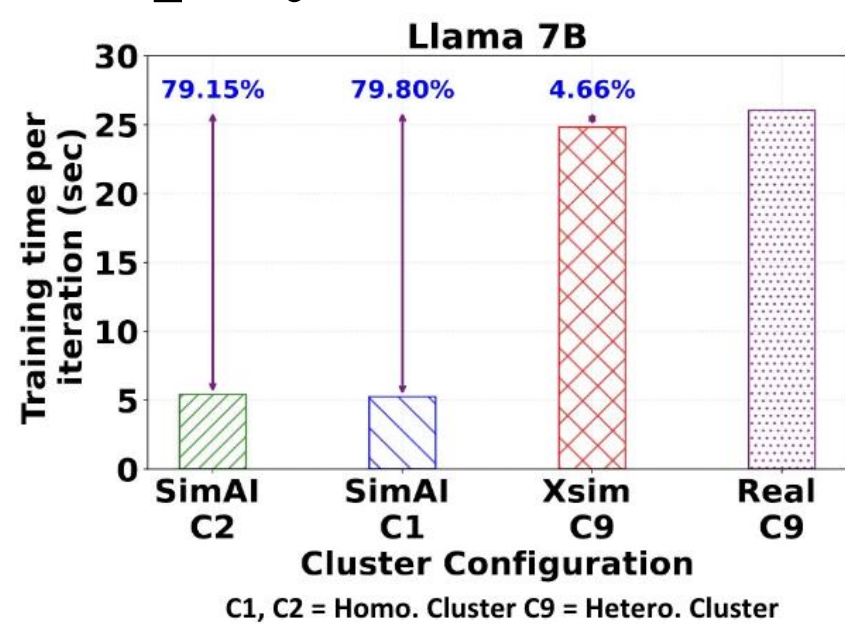
Distributed Heterogeneous ML Simulation is necessary

Challenges

Cluster is heterogeneous, but deployment is homogeneous



Cluster is homogeneous, but deployment is heterogeneous



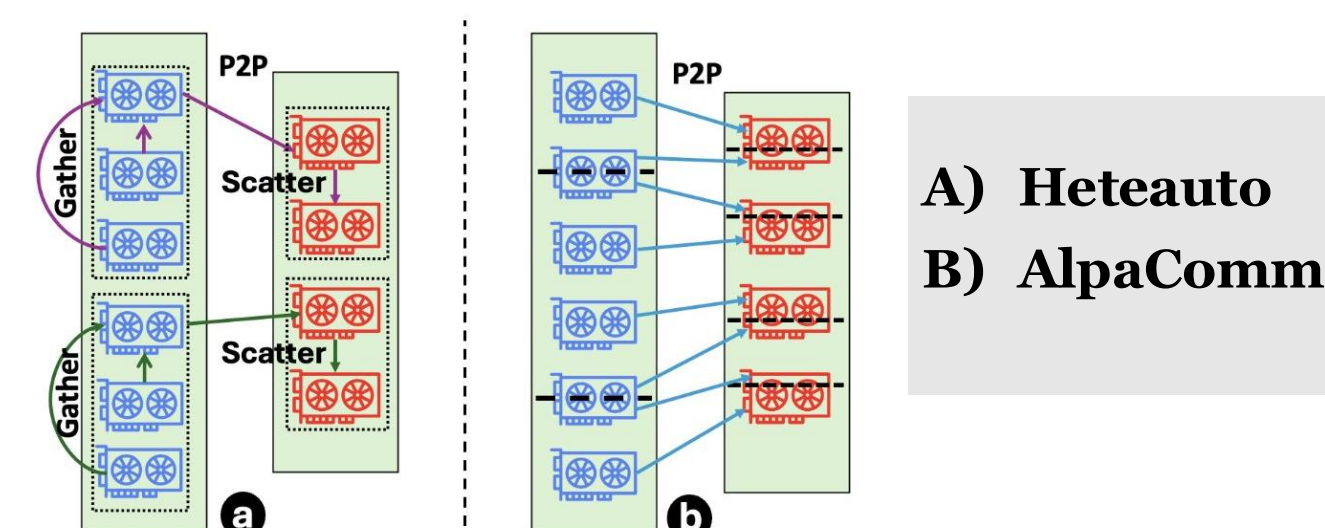
Related Work

Heterogeneity-aware deployment related work

	Non-Uniform deployment strategies	Validate using Real world	Validate using Analytical Simulation	Full Stack Simulation	Heterogeneous Communication
HeterMoe ^[1]	✓	✓	✗	✗	Send/Recv
Whale ^[2]	✓	✓	✗	✗	Bridge
Metis ^[3]	✓	✓	✗	✗	Centralized Resharding
HexScale ^[4]	✓	✓	✓	✗	Leader
Hetauto ^[5]	✓	✓	✗	✗	GCD-based Gather Scatter
AlpaComm ^[6]	✓	✓	✗	✗	Cutpoint-Union

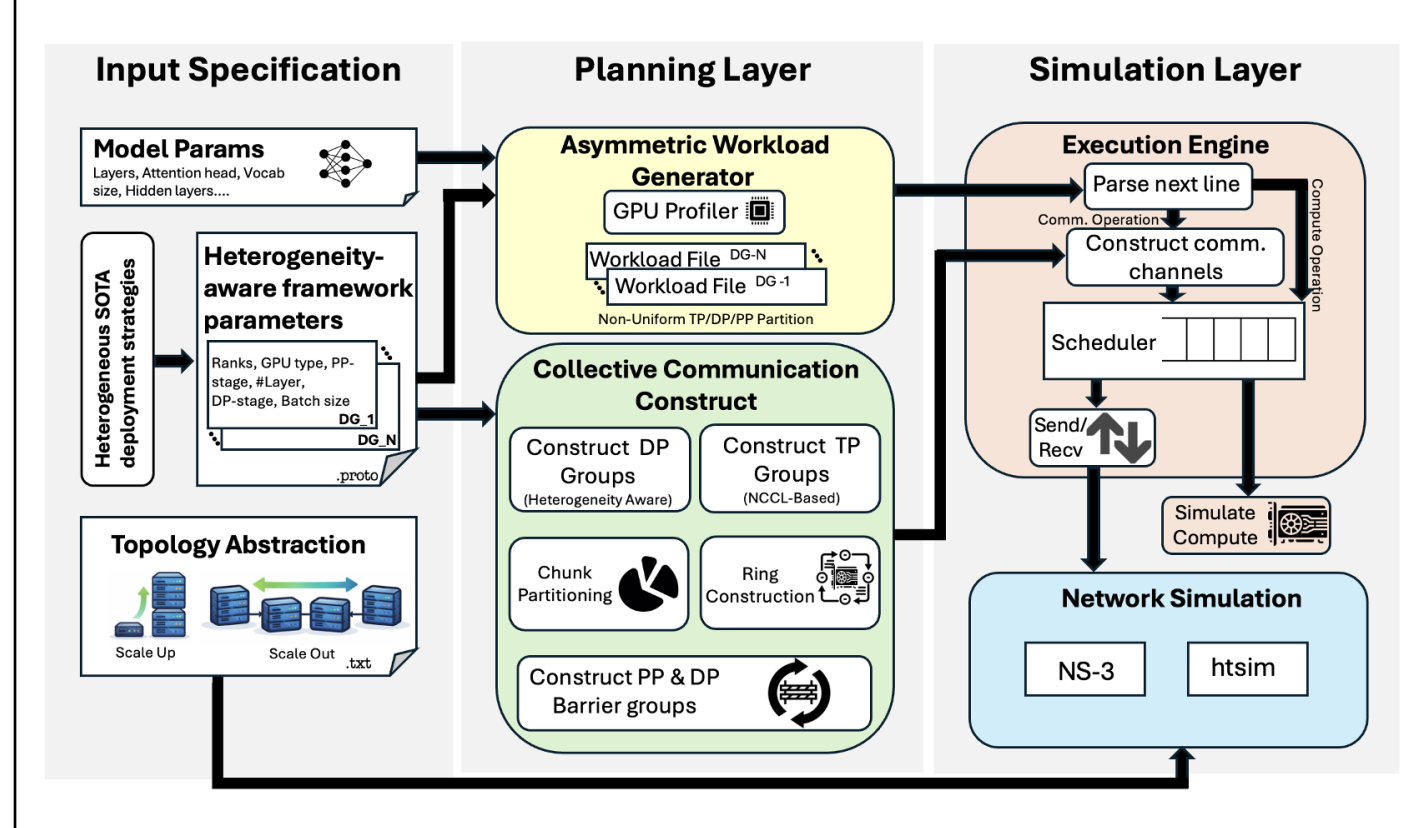
✗ Indicates no support ✓ indicates full support

SOTA Resharding Technique



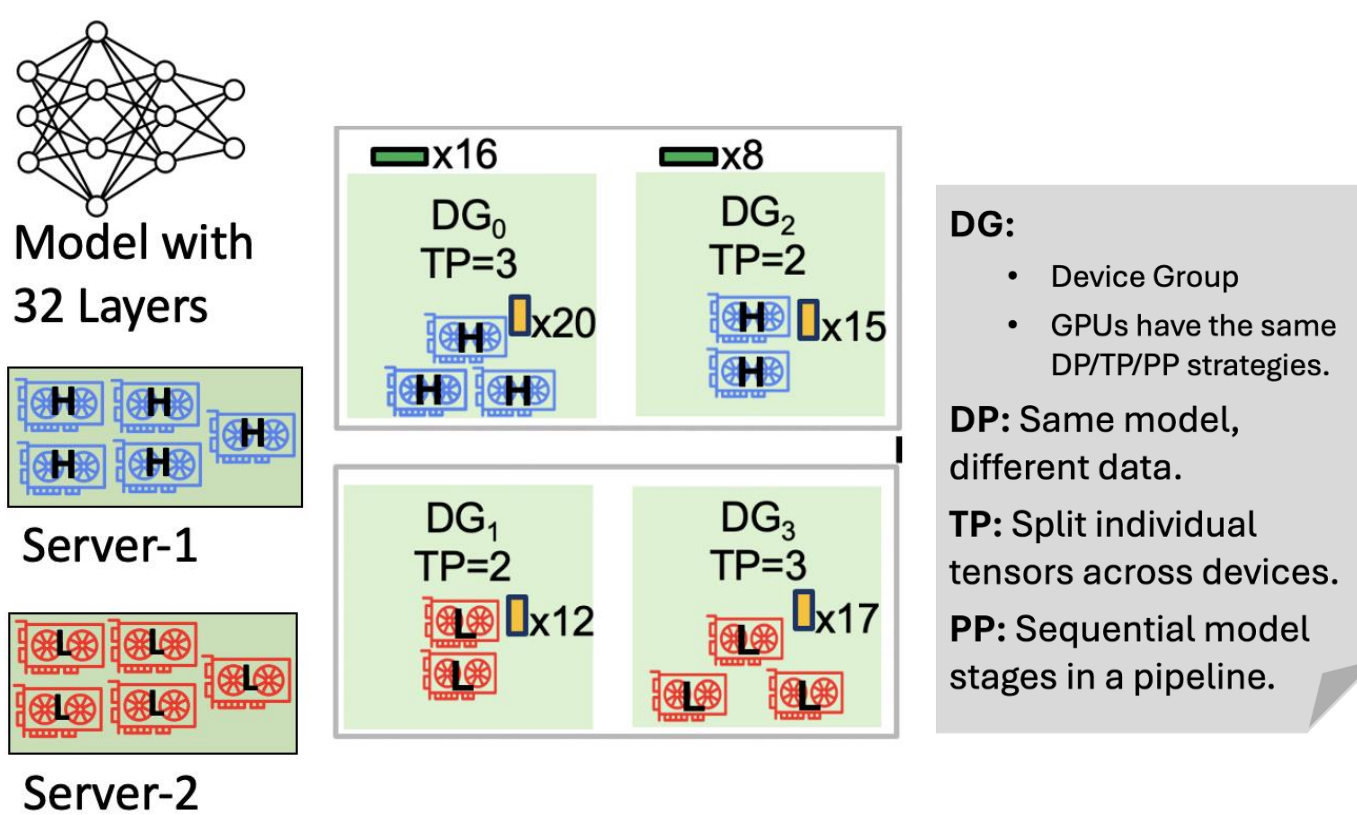
Only targeted for the P2P communication

Design

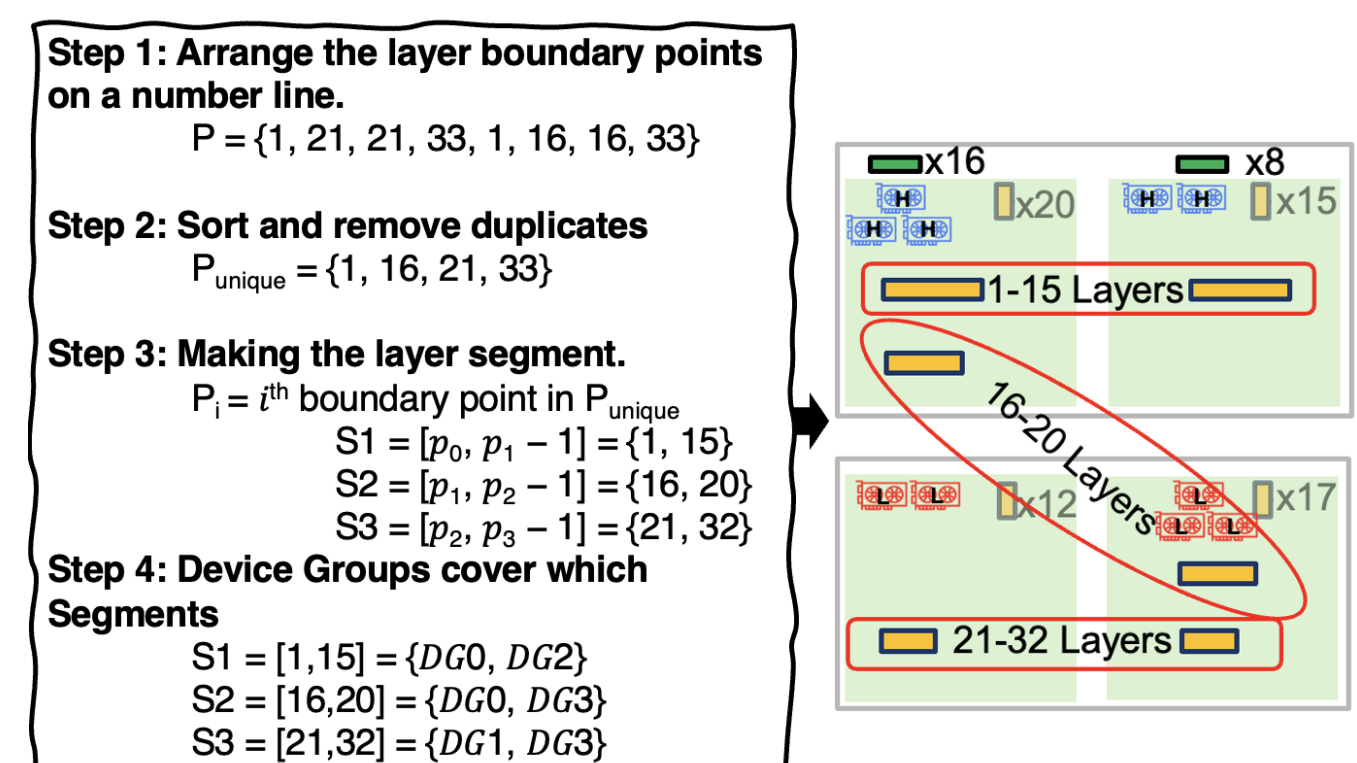


Methodology

Workflow of Heterogeneous Training

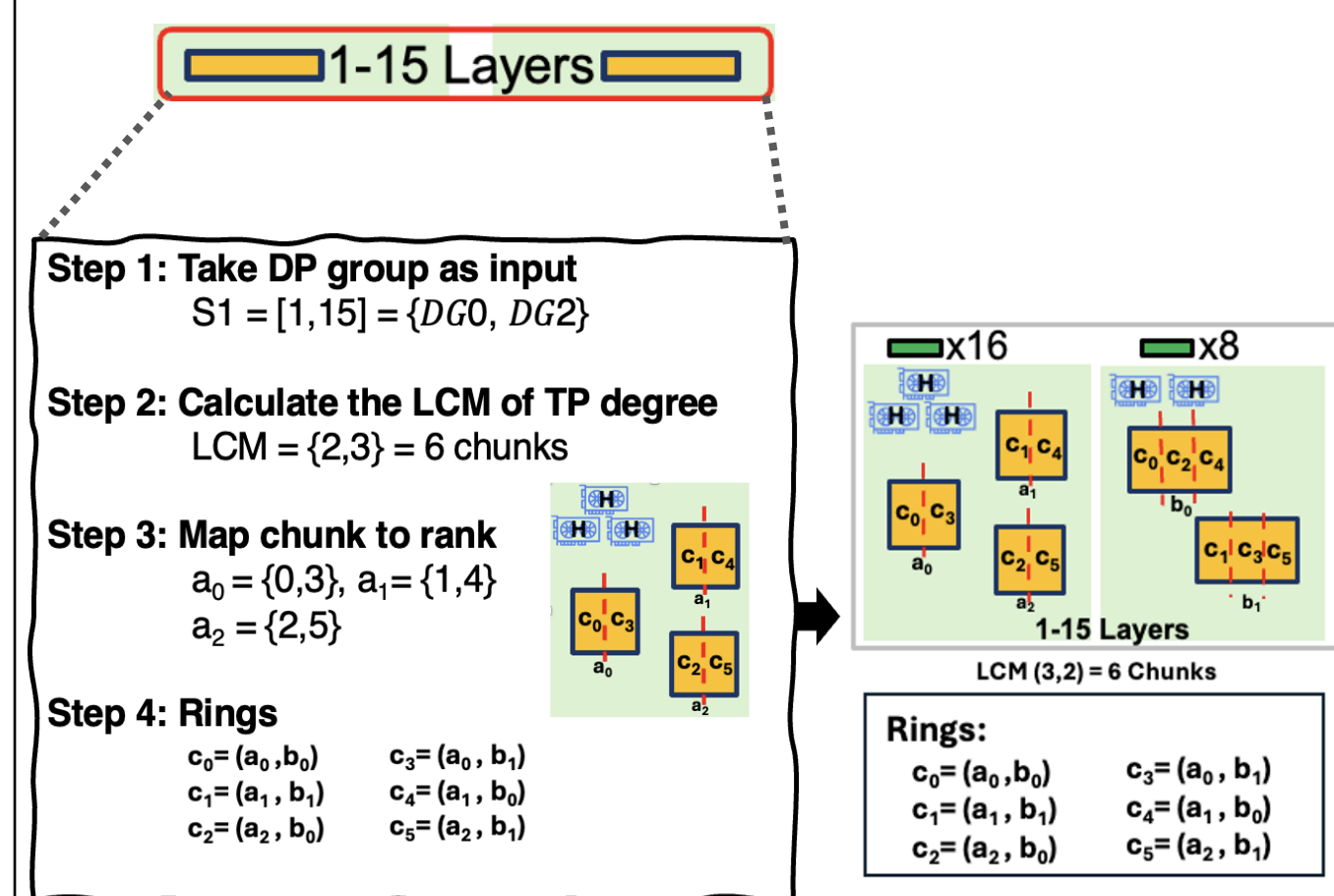


Heterogeneity-aware DP group Construction



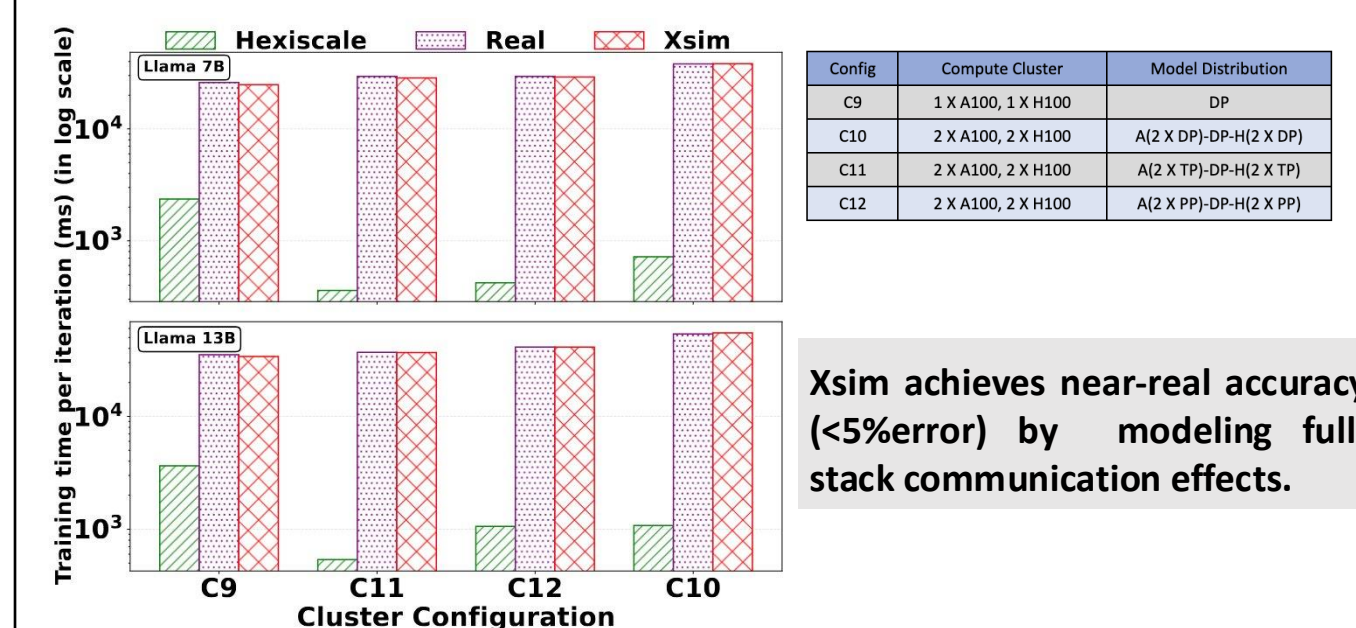
Methodology

Heterogeneity-aware ring Construction

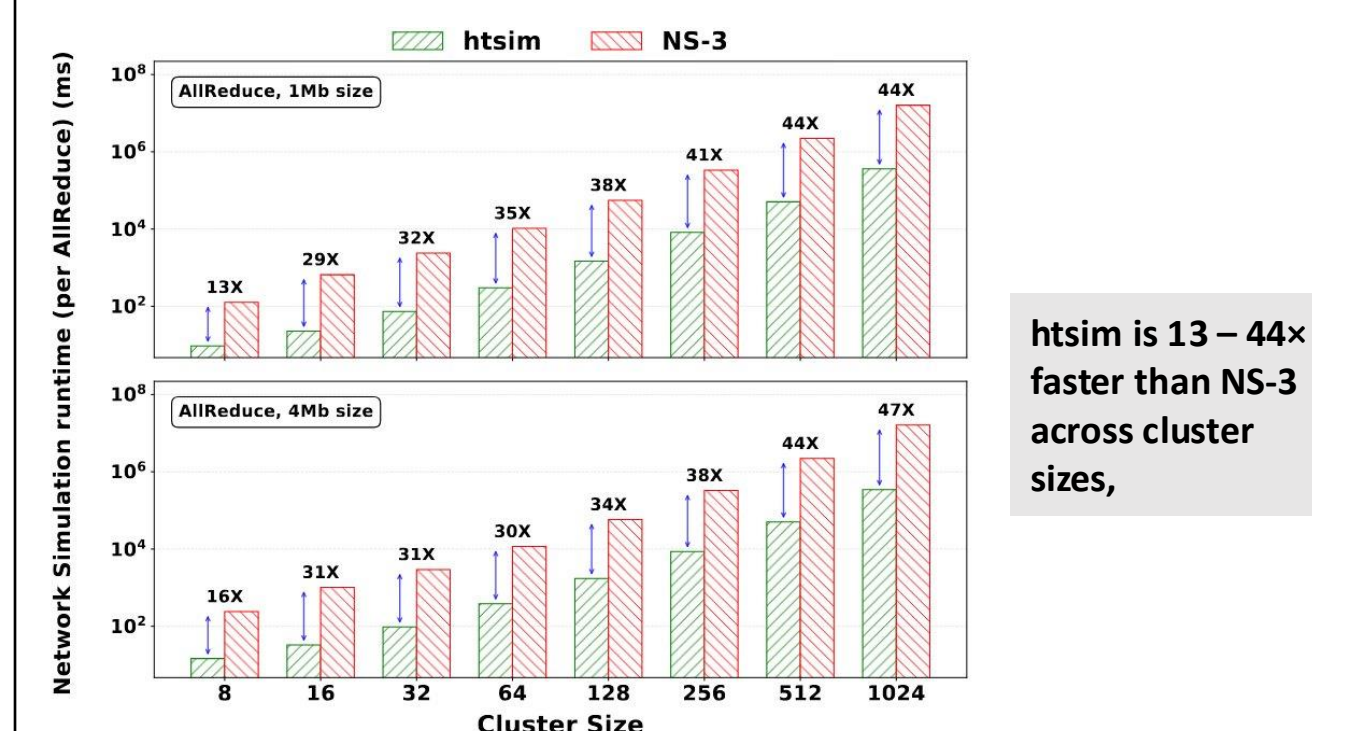


Results

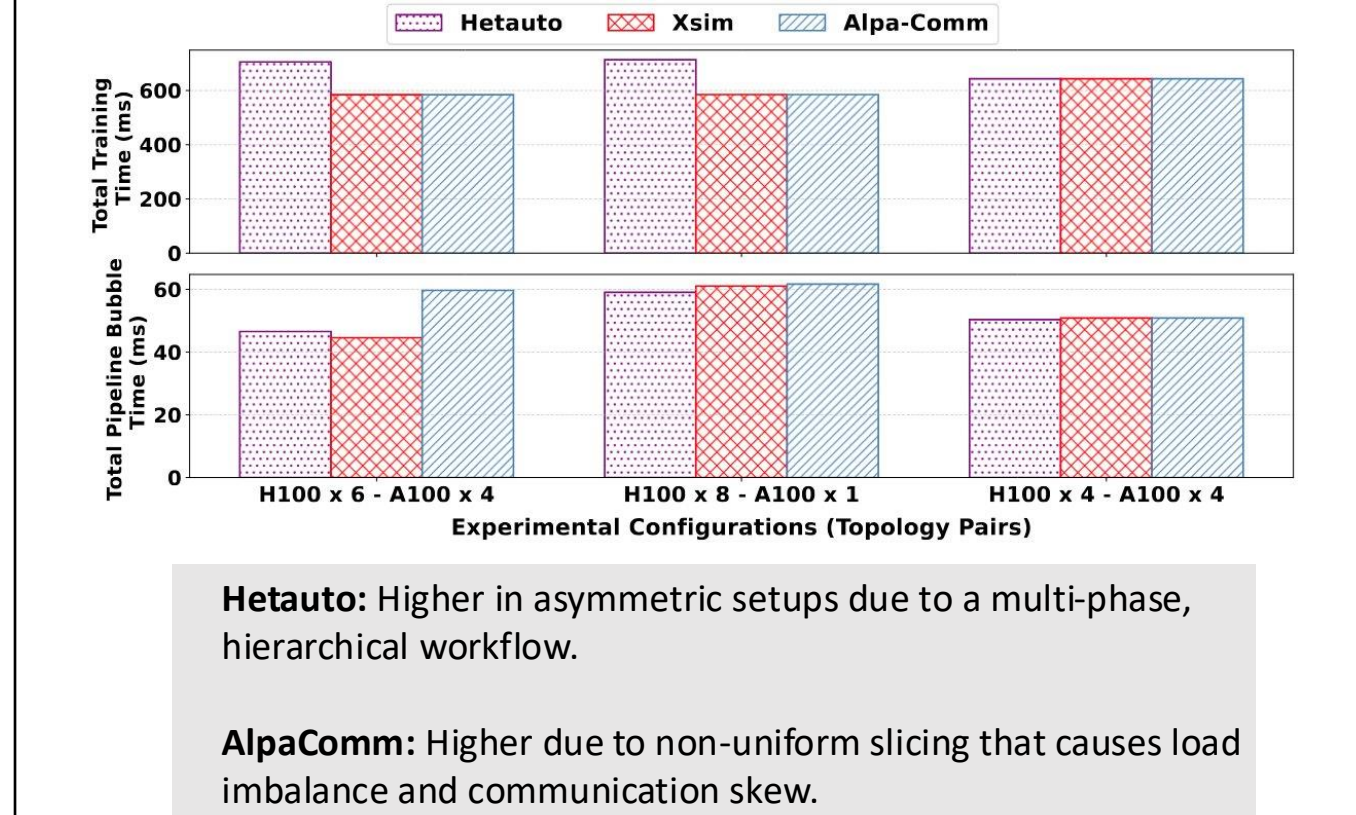
Accuracy



Scalability



SOTA Performance



Conclusion

- Accurate simulation of heterogeneous hybrid-parallel AI training across asymmetric hardware configurations using device groups, sweep-line DP formation, LCM-based synchronization, and gradient resharding.
- Equips researchers to systematically study co-evolving models, hardware, and parallelization techniques before deploying them at a massive scale.

References

- [1] HeterMoe: Efficient training of mixture-of-experts models on heterogeneous gpus
- [2] Whale: Efficient giant model training on heterogeneous (GPUs)
- [3] Metis: Fast automatic distributed training on heterogeneous (GPUs)
- [4] HexScale: Accommodating Large Language Model Training over Heterogeneous Environment
- [5] Hetauto: Cross-Cluster Auto-Parallelism for Heterogeneous Distributed Training
- [6] On optimizing the communication of model parallelism
- [7] Astra-sim2.0: Modeling hierarchical networks and disaggregated systems for large-model training at scale
- [8] (SimAI): Unifying Architecture Design and Performance Tuning for (Large-Scale) Large Language Model Training with Scalability and Precision